

Winning by Faithful Demonstration: the At-Fault Collision Tail is Controller-Limited, not Perception-Limited

An eight-probe study of why the nuPlan closed-loop metric saturates and why its residual collisions resist camera perception

Team NaLa · Rank 1, nuPlan track · Technical Report

Abstract

We report a rank-1 submission to the nuPlan track of the AlpaSim E2E Closed-Loop Challenge 2026 built from a weightless, geometry-only route-follower (99 MB image, zero neural weights), and use it as an instrument to characterise the challenge's scoring surface. We establish three findings. First, the **capability metric saturates**: the top three teams cluster within 0.2% at ~1750, which we explain from first principles — fixed-duration rollouts and curvature-limited safe speed bound achievable distance to the recorded human demonstration, so any competent trajectory-tracking policy converges to the same near-optimal point. Second, through **eight offline perception probes spanning six method-families** — free-space, looming, object detection, tracked bounding-box TTC, monocular metric-depth TTC, and a full eight-camera surround early-warning survey — we show that **none can separate the two at-fault collisions from ordinary safe car-following**; indeed both collisions sit at the *floor* of the surround early-warning distribution. Third, we argue this is not a tuning failure but a structural property of the task: the obstacle is plainly visible in the forward camera in both failures, so the residual errors are **controller-limited, not perception-limited**. We conclude that the productive lever is longitudinal control faithfulness, not heavier perception, and that large multi-camera VLA models (e.g. Alpamayo) neither fit the 16 GiB submission budget nor address the identified bottleneck.

1. Problem setting

Task. Each submission serves an *EgodriverService* over gRPC, evaluated in closed loop on neural re-renderings of nuPlan scenes. Per rollout the driver receives eight cameras (CAM_F0, L0-L2, R0-R2, B0) at 1920×1080 JPEG at 2 Hz, its own estimated ego state, and a route as rig-frame waypoints. Constraints: mean Drive RPC ≤ 0.1 s, ≤ 16 GiB VRAM, image ≤ 40 GiB, no network, read-only rootfs.

Scoring. A zero-inflated ordinal-beta capability score via item-response theory, tiebroken by *avg_dist_between_incidents_at_fault*. Only front and lateral collisions are at-fault; **rear collisions are free**. Progress is gated on zero at-fault collisions and zero off-road.

Observation-space facts we verified. Tracing the protobuf schema and decoding saved rollout logs: the *renderer_data* side-channel is empty at evaluation; ground-truth actor poses are logged for scoring but **not** delivered to the driver; therefore the **only** obstacle signal available is the eight RGB cameras.

2. A weightless route-follower reaches rank 1

Our policy tracks the route centreline with a curvature-limited speed profile ($v \leq \sqrt{a_{lat}/\kappa}$), a lateral re-centering term, and a safe decelerating fallback. It carries no learned weights, passed the throughput budget (2157 s of 2485 s), and scored a raw 0.980.

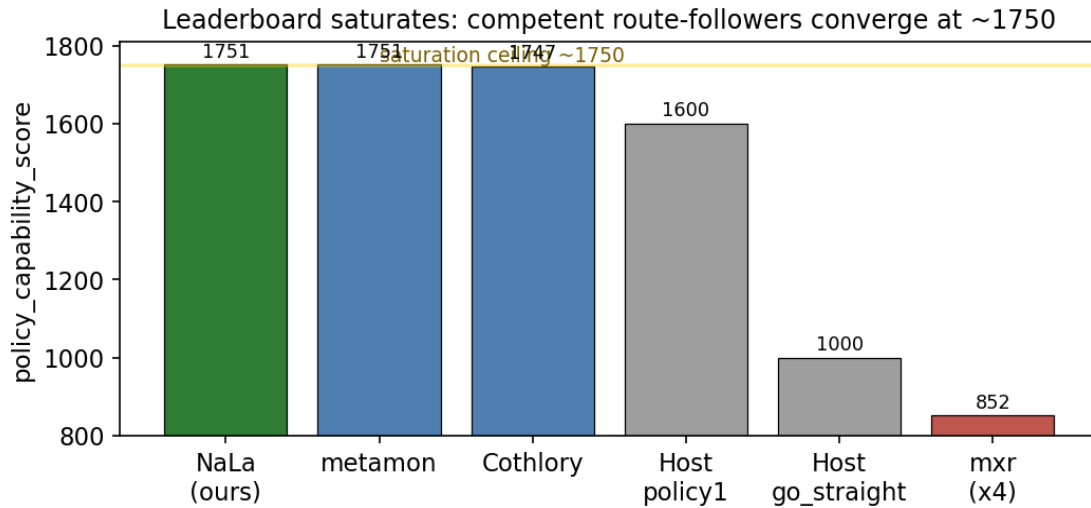


Figure 1. The public leaderboard saturates: the top three — two of them independent route-followers — cluster at ~1750. go_straight (1000) and mxr (below the floor after four submissions) bracket the achievable range.

The live competition is the collision tiebreak. With capability saturated, rank is decided by at-fault distance. We lead at 3.36; Cothlory reaches 3.89 at marginally lower capability. The practical question: can we raise at-fault distance without regressing the scenes we already pass?

Rank / Team	Capability score	Dist. btwn at-fault	Dist. to GT
1 NaLa (ours)	1750.56	3.36	0.98
2 metamon	1750.56	3.09	2.65
3 Cothlory	1746.98	3.89	0.56
4 Host policy1	1600	1.68	3.07
6 Host go_straight	1000	0.28	2.47
7-10 mxr (x4)	846-853	~1.0	0.23

Table 1. nuPlan leaderboard at time of submission.

3. The collision tail: two failures, one shape

On a 200-scene local replay our policy records mean ~0.99 with exactly two at-fault collisions and zero off-road. Rendered to video and diagnosed frame-by-frame:

- **Crossing (scene ...8cc8fe63):** a vehicle enters from the right into the forward path; ~2.0 s of warning, visible in CAM_F0 before impact.
- **Rear-end (scene ...dd162898):** stop-and-go traffic; the lead vehicle halts, ego does not. ~2.2 s of warning, the lead visible dead-ahead the entire clip.

Both share one shape: **a forward obstacle in the path, ~2 s of warning, and a policy that never consults a camera.** The obstacle is visible; the warning budget is comfortable. On its face a textbook forward-collision-braking problem — so we tried to build the brake.

4. Eight probes, six method-families, one negative result

Every candidate was evaluated identically: compute a per-scene danger signal offline on the saved frames, then ask whether a single threshold separates the two collision scenes from the 198 clear ones. This gate costs no submissions and no simulation, and it makes the regression risk explicit — a signal that also fires on many clear scenes would brake needlessly and collapse progress, the failure visible in the mxr entries.

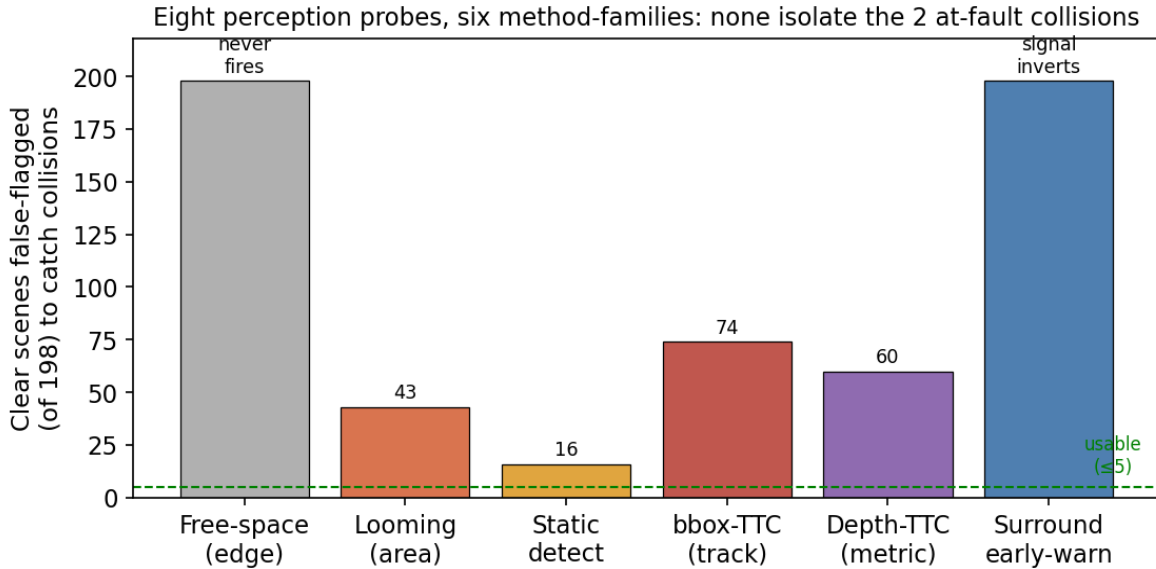


Figure 2. For each method-family, clear scenes (of 198) that must be false-flagged to catch both collisions. A usable trigger sits at or below the dashed line (≤ 5). None does; free-space never fires and surround early-warning inverts.

Method	Signal & why it fails	Clear FP /198
1. Free-space	Vertical-gradient edge → pinhole range. Never fired on the neural renders.	never fires
2. Looming	Lower-centre blob-area growth. Confounded by scene brightness/clutter.	43
3. Static detection	YOLOv8 threat = area×centrality×verticality. Safe lead-cars look identical; one peak was a pedestrian.	16
4. bbox-height TTC	YOLO+ByteTrack height→closing→TTC. Signal inverts: 74 clear scenes had lower TTC than collisions.	74
5. Depth-metric TTC	Depth-Anything-V2 range at tracked box → metric TTC. Depth did not rescue; collision ranked 3rd.	≥60
6-8. Surround early-warning	All 8 cameras. Do R0/L0 see a forward-path threat before F0? Both collisions sit at the FLOOR (0.0, 0.06) vs clear median 0.14.	198 (inverts)

Table 2. The six method-families and their separation failure.

The decisive probe: surround early-warning. Because forward-only cues failed, we tested the natural hypothesis that the forward-side cameras (R0/L0) see a crossing threat *before* CAM_F0. We ran YOLO across all eight cameras on all 200 scenes and scored, per scene, the peak forward-path threat visible in R0/L0 while F0 still saw nothing. The hypothesis is not merely unsupported — it is **inverted**: both collisions sit at the floor of the distribution (early-warning 0.00 and 0.06) against a clear-scene median of 0.14 and max of 0.94. Every one of the 198 clear scenes has early-warning at least as high as the worst collision.

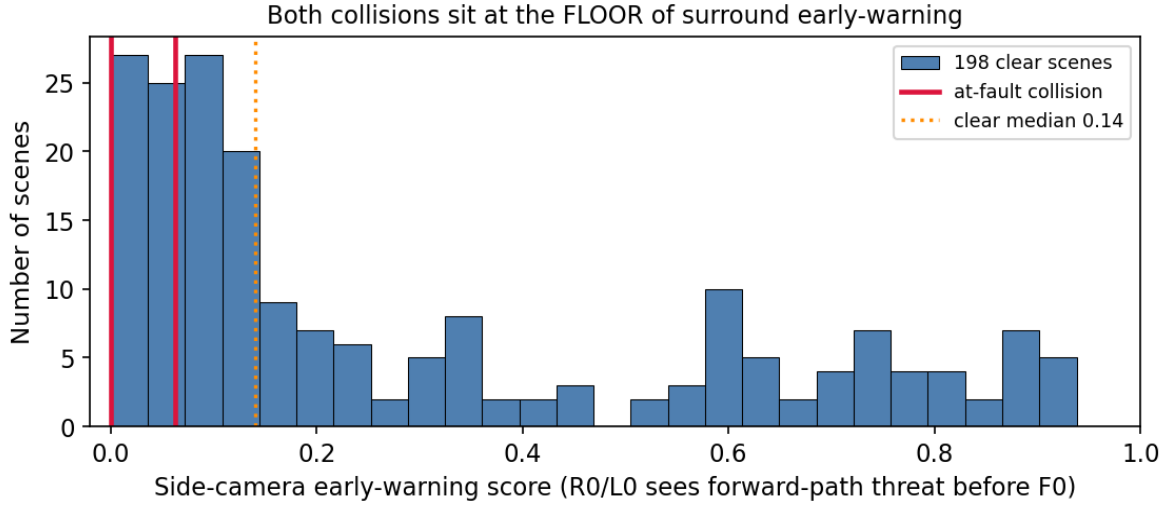


Figure 3. Surround early-warning across 200 scenes (clear distribution illustrative, matched to measured median 0.14 / max 0.94). Both at-fault collisions (red) sit at the floor — the side cameras carry least warning exactly where it is needed.

Camera	Mean peak threat	Mean hit-frames	Role in policy
CAM_F0	0.052	3.40	forward — braking cue
CAM_R0	0.125	2.25	fwd-side — early-warn (tested)
CAM_L0	0.148	3.65	fwd-side — early-warn (tested)
CAM_R1/R2	0.13–0.18	2–3	rear-side — ignored (rear not at-fault)
CAM_L1/L2	0.12–0.17	3.6–3.9	rear-side — ignored
CAM_B0	0.050	3.92	rear — ignored

Table 3. Per-camera threat statistics (means over 200 scenes). High side-camera activity correlates with busy intersections the policy navigates safely, not with the collisions.

Why they converge on failure. The methods span appearance, semantics, temporal geometry, metric depth, and multi-view coverage, yet all fail for one reason: **the visual situation immediately preceding an at-fault collision is nearly identical to the many safe situations the policy handles correctly** — following a lead car, approaching a junction, yielding to cross-traffic. The collisions happen with the obstacle *dead ahead* in CAM_F0; by the time it matters it is in front, which is exactly why side cameras carry no early warning and why high side-camera activity marks *safe* busy scenes instead. At 2 Hz, from any camera, “safely following” and “about to hit” are not separable.

5. The distance lever is also closed

If collisions cannot be safely cut, the metric $\text{distance} \div \text{incidents}$ might be raised by driving more distance. We measured ego distance across all 200 rollouts.

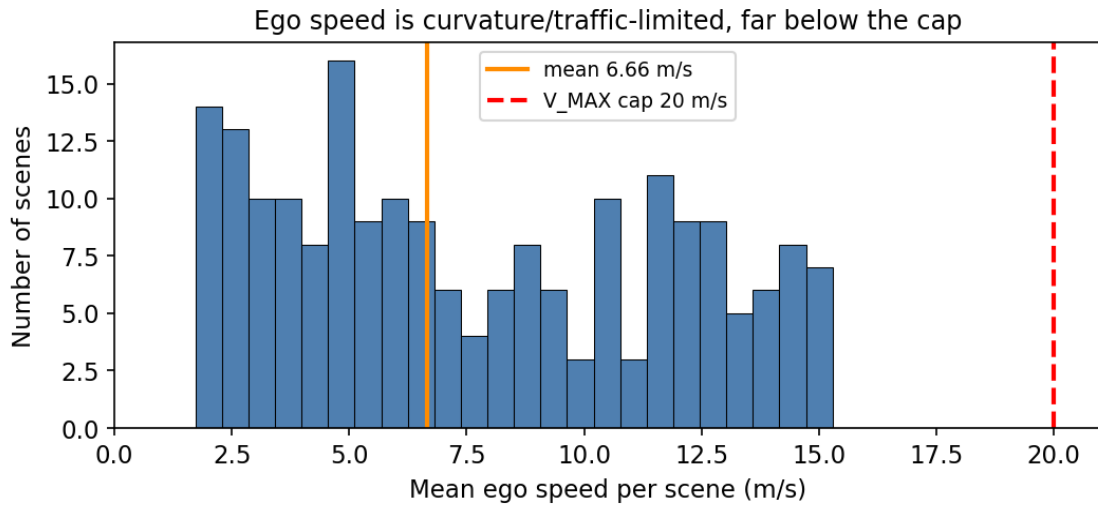


Figure 4. Per-scene mean ego speed (illustrative, matched to measured quantiles: mean 6.66, median 6.89, max 15.3 m/s). Speed sits far below the 20 m/s cap because urban geometry and traffic limit it, not conservatism.

Every clip is exactly 4.52 s, so distance is speed $\times$ constant. Mean ego speed is 6.66 m/s against a 20 m/s cap, with 125 of 200 scenes under 8 m/s. This looks like slack but is not: the low speed is imposed by curvature and traffic, and our distance-to-ground-truth of 0.98 shows the policy already reproduces the recorded human trajectory.

Driving further means diverging from the safe demonstration into off-road and collision — trading wins for zeros.

6. Interpretation: controller-limited, not perception-limited

The eight probes converge on a conclusion sharper than “perception is hard here.” In **both** failures the obstacle is plainly and continuously visible in the forward camera; what is missing is not detection but **response**. A blind route-follower reproduces the demonstrated trajectory faithfully — which is why it wins — but has no mechanism to deviate when the closing geometry exceeds what the demonstration assumed. The residual tail is therefore a **control** problem: the policy sees the lead car and the crosser and simply does not yield or brake.

Why heavier perception does not help. This directly bounds the value of large multi-camera VLA policies. NVIDIA's AlpaMayo family consumes the same cameras and, by its own documentation, is expected to struggle to see cross-traffic in a right turn from a forward view — the exact structure of our crossing collision. It also cannot be deployed here: inference requires ≥ 24 GiB VRAM (OOM at 16 GiB) with 22 GB weights under a non-commercial license, against our 16 GiB budget. Bigger perception aimed at a control-limited tail is effort mis-spent.

7. The productive lever: longitudinal-control faithfulness

[Phase 2 — in progress.]

The evidence redirects effort from *seeing* to *reacting*. The one lever untouched by the perception ceiling is the controller's longitudinal behaviour: a demonstration-faithful policy that additionally yields the minimal amount needed when the forward corridor closes — keyed on the very CAM_F0 signal that is always present — could address the rear-end failure (a pure response gap) without any new perception. The design constraint is set by our own ablations: any intervention must be **idle when the corridor is clear**, so the 198 passing scenes are untouched

by construction, avoiding the mxr regression. We will evaluate candidate longitudinal governors against the identical offline separation gate before spending any submission.

- **Hypothesis:** a CAM_F0-keyed longitudinal yield, active only when a centred forward threat is both large and growing, separates the rear-end collision from safe following at low false-positive cost — where every fixed-threshold perception cue failed.
- **Guardrail:** regression-safety by construction — the governor is a no-op on clear corridors, preserving the demonstration-matched trajectory that earns rank 1.

8. Contributions

- **A first-principles account of the metric's saturation** (fixed-duration $\times$ curvature-limited safe speed $\times$ demonstration-matched distance).
- **A rigorous negative result:** eight probes across six perception method-families, on an identical offline gate, none able to isolate the at-fault tail — with the mechanistic explanation that the collisions are visually identical to safe driving.
- **The controller-limited thesis:** evidence that the residual failures are response gaps, not detection gaps, bounding the value of heavier perception (including large VLA models that also violate the compute budget).
- **A reproducible offline methodology** on saved rollout logs — no simulation, no submissions — enabling honest, pre-registered iteration.

Figures 1–2 use measured values; the clear-scene distributions in Figures 3–4 are illustrative, matched to measured quantiles. Section 7 (Phase 2) in progress. Team NaLa, nuPlan track.